

- 1) Given: V is a finite dimension vector space over field F .

Let $T \in L(V)$.

Let B be any basis of V .

$[T]_B$ is matrix representation of T wrt B .

To prove:

- a) T is invertible iff $[T]_B$ is invertible.
b) $\lambda \in F$. λ is eigenvalue of T iff matrix $[T - \lambda I]_B$ isn't invertible.

Proof:

- a) Let $n = \dim V$.

forward proof:

Assuming T is invertible

$$\exists T^{-1} \in L(V), \text{ st. } T \cdot T^{-1} = I \\ \text{and } T^{-1} \cdot T = I.$$

taking basis B ,

$$[T T^{-1}]_B = [T]_B [T^{-1}]_B = I_B = I_n.$$

$$[T^{-1} T]_B = [T^{-1}]_B [T]_B = I_n.$$

So, T_B has inverse, $[T^{-1}]_B$. So

$[T]_B$ is invertible.

backward:

Let $[T]_B$ is invertible.

$$\therefore \exists A \text{ st } [T]_B A = A [T]_B^{-1} = I_n$$

Let $S \in L(V)$
linear operator.

$$[S]_B = A.$$

$$\text{Then: } [T, S]_B = [T]_B [S]_B = [I]_B$$

$$[ST]_B = [S]_B [T]_B = [I]_B$$

$$\therefore TS = ST = I$$

$$\therefore S = T^{-1}$$

$\therefore T$ is invertible.

Hence proved.

Hence proved.

2) Given: for V over field K ,
let $T \in L(V)$ be
invertible.

To prove: $E(\lambda, T) = E(\frac{1}{\lambda}, T^{-1})$

for $\forall \lambda \in K, \lambda \neq 0$.

Proof:

let $v \in E(\lambda, T)$

$$v \neq 0$$

$$\therefore T(v) = \lambda v.$$

Since T is invertible,

$$T^{-1}(T(v)) = T^{-1}(\lambda v)$$

$$v = \lambda(T^{-1}(v))$$

$$\text{So, } \frac{v}{\lambda} = T^{-1}(v).$$

$$\therefore v \in E\left(\frac{1}{\lambda}, T^{-1}\right)$$

$$\therefore E(\lambda, T) \subseteq E\left(\frac{1}{\lambda}, T^{-1}\right) \quad (1)$$

b). forward proof:

Let λ is eigenvalue of T .

$$\therefore \exists v \neq 0, T(v) = \lambda v.$$

$$\therefore (T - \lambda I)v = 0.$$

So, $T - \lambda I$ is NON-TRIVIAL Kernel.

$\therefore [T - \lambda I]$ isn't invertible.

Using (a), $[T - \lambda I]_B$ isn't invertible.

backward proof:

Let $[T - \lambda I]$ isn't invertible

$$\therefore \exists v \neq 0,$$

$$[T - \lambda I](v) = 0.$$

$$T(v) = \lambda v$$

$$\therefore T(v) = \lambda v$$

$\therefore \lambda$ is eigenvalue & v is

Now, Let $v \in E\left(\frac{1}{\lambda}, \tau''\right)$, $v \neq 0$.

$$\text{so, } T^{-1}(v) = \frac{1}{\lambda} v.$$

Applying T on both sides,

$$T(T^{-1}(v)) = T\left(\frac{1}{\lambda} v\right)$$

$$v = \frac{1}{\lambda} (T(v))$$

$$\boxed{\lambda v = T(v)}$$

$$\text{so, } v \in E(\lambda, T)$$

$$\therefore \boxed{E\left(\frac{1}{\lambda}, \tau''\right) \subseteq E(\lambda, T)} \quad \text{--- (2)}$$

so, from (1) & (2),

$$\boxed{E(\lambda, T) = E\left(\frac{1}{\lambda}, \tau''\right)}$$

Date: _____ YOUVA

3). Suppose matrix of linear operator T , relative to standard basis, is given by:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 12 \end{bmatrix}$$

Can it be diagonalised?

To find: Basis rel to which T is diagonal

Solⁿ: Since A is upper triangle matrix, eigenvalues are diagonal ~~at~~ entries:

$$\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 12.$$

A matrix with n distinct eigenvalues is diagonalizable.

$\therefore A$ is diagonalizable.

for $\lambda = 1$,
 $(A - I)x = 0$

$$\therefore A - I = \begin{bmatrix} 0 & 2 & 1 \\ 0 & 1 & 4 \\ 0 & 0 & 11 \end{bmatrix}$$

$\therefore$ ~~$0x$~~ $2y + z = 0$
 $y + 4z = 0$
 $11z = 0$

Then: $z = 0, y = 0, x$ is free

$\therefore$ eigenvector for $\lambda_1 = 1$ is $(1, 0, 0)$

for $\lambda = 2$,

$(A - 2I)x = 0$

$$A - 2I = \begin{bmatrix} -1 & 2 & 1 \\ 0 & 0 & 4 \\ 0 & 0 & 10 \end{bmatrix}$$

$\therefore 10z = 0$

$4z = 0$

$-x + 2y + 4z = 0$

$\therefore z = 0$

$2y = x$

$$\therefore \text{if } y = 1, z = 2,$$

$$\therefore v_2 = (2, 1, 0).$$

for $\lambda = 12$,

$$(A - 12I)z = 0.$$

$$\therefore A - 12I = \begin{bmatrix} -11 & 2 & 1 \\ 0 & -10 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore -10y + 4z = 0$$

$$-10y + 4z = 0$$

$$2z = 5y$$

$$2z = 5y$$

$$\text{if } \boxed{z = 55}$$

$$2 \cdot 55 = 110$$

$$\boxed{y = 22}$$

$$-12x + 99 + 110 = 0$$

$$-11x + 2y + z = 0.$$

$$-11x + 99 + 55 = 0$$

$$+11x = -99 - 55$$

$$x = -9.$$

$$\therefore v_3 = (9, 22, 55)$$

so, 3 linearly independent
eigenvectors:

$$\{(1, 0, 0), (2, 1, 0), (9, 22, 55)\}$$

form basis of eigenvectors, so T is
diagonalizable.

In basis, matrix of T is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 12 \end{bmatrix}$$